

Rajan Patel

+91 7009853307

rkp1505.l@gmail.com

LinkedIn

Github

Portfolio

Education

I.K.G. Punjab Technical University

Bachelor of Technology in Computer Science

Aug. 2023 – June 2027

CGPA: 8.50/10

Experience

Small Fare™

Dec 2025 – April 2026

Backend Developer Intern | Node.js, PostgreSQL, Prisma, Cassandra, Redis, Kafka, Docker

Hyderabad, Telangana

- Engineered an **event-driven Kafka pipeline** (idempotent producer, exactly-once semantics, GZIP) decoupling booking, ticketing, and search indexing.
- Architected dual-database sync: **Cassandra for high-throughput TTL** ticket holds and **PostgreSQL for ACID transactions**, synced via drift-detection workers.
- Built a fault-tolerant **webhook** engine using **Redis distributed locks** (atomic Lua unlocks), 24h replay protection, and **HMAC-verified** confirmations.

Wyreflow Technologies

July 2025 – Oct 2025

Full Stack Developer Intern | Node.js, Express, MongoDB, React.js

Bhopal, Madhya Pradesh

- Developed a 3-tier **RBAC** system (Student, Sub-Admin, Super Admin) with custom **JWT middleware** and per-module HTTP method permission guards.
- Integrated Razorpay for memberships and bookings, implementing HMAC-SHA256 signature verification and webhook-driven order lifecycle handling.
- Built the end-to-end job application workflow with real-time admin notifications (Node EventEmitter) and deadline-validated, state-tracked application status transitions.

Projects

State University - Alumni Network Platform | [Demo](#)

July 2026

Next.js, TypeScript, PostgreSQL, Prisma, OAuth, Redis, Leaflet.js, Cloudinary, Vercel, Brevo

- Architected a multi-tenant portal scaling across **6 campuses & 250+ colleges**, isolating data across **30+ relational models** via indexed foreign keys.
- Built a **FK-based data normalization engine** that collapsed **40+ branch & 250+ company variants** across **5000+ records** via atomic Prisma transactions, raising filter precision to **100%**.
- Built a **keyset-paginated** notification worker processing **10K notifications in 2.39s**, plus a rate-throttled SMTP pipeline with zero bounces.

InVolv IN | [Github](#) | [Demo](#)

December 2025

MERN, OAuth, Redis, Leaflet.js, Zustand, Firebase CM, Docker, Cloudinary, OSRM, ML/FastAPI(statsmodel), CronJob

- Built a **geospatial discovery engine** using MongoDB **\$geoNear**, Atlas fuzzy **\$search** for real-time vendor matching.
- Shipped an **ML demand-forecasting service** (FastAPI, **SimpleExpSmoothing**) with **Redis-cached (24h)** responses for price-trend prediction.
- Built a real-time analytics pipeline (**Redis Hashes/Sets**) and cronjob for automated restock **FCM** alerts.

Onwards AI- Voice-Guided Browser Navigation Agent | [Github](#)

May 2026

React, Node.js, Express Groq/Gemini API, WebSpeech API, html2canvas

- Built a **voice-guided AI browser agent** combining **WebSpeech API**, **Groq Llama 3.3** LLM inference, and deterministic DOM automation.
- Cut LLM payload to **70%** by two-stage **token-budget pipeline** (DOM purification + 24K-char backend truncation).
- Eliminated selector hallucinations to **0%** using a deterministic data-unstuck-id DOM-tagging & routing engine.

Technical Skills

Programming Languages: C, C++, JavaScript, TypeScript, Python, SQL

Frameworks & Techs: React.js, Next.js, Node.js, Express.js, REST APIs, Webhooks, Cron Jobs, Zustand

Databases & ORM: MySQL, PostgreSQL, MongoDB, Cassandra DB, Redis, Prisma ORM

Tools & Cloud: AWS, Git, Docker, Kafka, Firebase, Cloudinary, Vercel, Render, Postman

AI/ML & Data: Machine Learning, LLM Integration, Data Analytics, Scikit-learn, Pandas, NumPy

Achievements

- Solved **450+** Data Structures & Algorithms problems on platforms like LeetCode
- Active University's **T&P Cell Coordinator** and **E-Cell** Core Member, managing **portal maintenance** and startup initiatives.
- Completed Training in **Machine Learning** via **IIT Madras Pravartak**, focusing on ML modeling.
- Finalist **HackMol 6.0** and **PSB Hackathon-2025** for building high-impact Products.